

Do Quantitative Easing and Quantitative Tightening Have Asymmetric Effects on Bond and Equity Risk Premia in Europe?

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M2 Economics of Global Risks – Empirical Project

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Motivation

- ▶ ECB balance sheet: from €2tn (2014) to €9tn (2022), now shrinking under QT
- ▶ **Policy question:** Is QT simply QE in reverse, or does the unwinding generate disproportionate financial tightening?
- ▶ Two channels suggest asymmetry:
 - **Signalling channel:** effective lower bound constrains rate expectations during QE, not during QT (Lloyd & Ostry, 2024)
 - **Fragmentation channel:** ECB purchases act as implicit backstop for peripheral sovereign debt; withdrawal may trigger non-linear spread widening
- ▶ **Gap in the literature:** existing evidence is US-focused (Treasury yields only); no study examines QE/QT asymmetry on euro-area *sovereign spreads* and *sectoral equity risk premia*

Research Question

Do QE and QT have asymmetric effects on sovereign bond spreads and equity risk premia in the euro area?

Three contributions:

1. Extend Lloyd & Ostry (2024) QE/QT asymmetry analysis from US Treasuries to **euro-area sovereign spreads** and **sectoral equity risk premia**
2. Combine Altavilla et al. (2019) high-frequency identification with **panel local projections** (Jordà, 2005) pooled across countries and sectors
3. Test the **equity-duration hypothesis**: do growth sectors react more than value sectors to balance-sheet shocks?

Data & Shock Construction

Sample: September 2004 to September 2025 (253 monthly observations)

Variables: Bond market data; Equity market data; Monetary policy measures; Control variables (86 in total)

Shock identification:

- ▶ QE/QT factor from the EA-MPD (Altavilla et al., 2019): rotated PCA on high-frequency OIS changes in ± 10 min window around ECB **press releases**
- ▶ Captures unexpected changes in ECB balance-sheet policy intentions

Regime classification:

Regime	Periods	Obs $\neq 0$
QE (early)	CBPP1, SMP, CBPP2, ABSPP/CBPP3 (2009–2014)	46
QE (APP)	APP + PEPP (2015–2022)	78
QT	Reinvestment end (June 2022 onward)	40

$$QE_shock_t = D_t^{QE} \times qe_release_m_t \quad QT_shock_t = D_t^{QT} \times qe_release_m_t$$

$$QE_Full_shock_t = (D_t^{QE} + D_t^{QE_early}) \times qe_release_m_t$$

Outcome Variables

Sovereign bond risk premia:

$$\text{Spread}_t^c \equiv y_{10y,t}^c - y_{10y,t}^{GE}, \quad c \in \{\text{IT, SP, PT, FR}\}$$

Where $y_{10y,t}^c$ denotes the 10-year government bond yield for country c .

Equity risk premia:

$$\text{ERP}_t^s \equiv r_t^s - r_t^f, \quad s \in \{\text{Tech, Utilities, Banks, Oil\&Gas}\}$$

Where r_t^s is the monthly return on sector s and $r_t^f = ((1 + r_{ECB}^f/100)^{1/12} - 1)$ is the monthly risk-free rate derived from the annualised ECB policy rate.

Controls:

- ▶ MP_t^{short} : 2-year OIS rate (conventional policy)
- ▶ Lagged inflation (log-diff HICP), ECB policy rate
- ▶ Entity fixed effects (country or sector)

Econometric Framework: Panel Local Projections

Panel LP at horizon $h \in \{0, 1, 3, 6, 12\}$ months (Jordà, 2005):

$$y_{i,t+h} - y_{i,t-1} = \alpha_i^h + \beta_{QE}^h QE_t + \beta_{QT}^h QT_t + \theta^h MP_t^{short} + \Gamma^{h'} X_{t-1} + \varepsilon_{i,t}^h$$

- ▶ **Pooling:** 4 countries (spreads) or 4 sectors (ERP) $\Rightarrow N \approx 830$
- ▶ **Fixed effects α_i^h :** absorb time-invariant heterogeneity across entities
- ▶ **Inference:** Newey-West HAC standard errors (6 lags)

Asymmetry test at each horizon:

$$H_0 : \beta_{QE}^h = \beta_{QT}^h \quad (\text{Wald statistic})$$

Results: Sovereign Spreads

Baseline specification (QE_Full_shock)

h	$\hat{\beta}_{QE}^h$	$\hat{\beta}_{QT}^h$	Wald p
0	-0.060*	+0.004	0.058*
1	-0.022	+0.017***	0.273
3	+0.005	+0.029***	0.608
6	+0.045	+0.026**	0.764
12	+0.092	+0.035**	0.368

Key findings:

- ▶ QT: persistent spread widening from $h = 1$ to $h = 12$ (all significant)
- ▶ QE: transitory compression at impact, insignificant thereafter
- ▶ Asymmetry in **persistence**, not just magnitude

Results: Peripheral vs. Core Sensitivity

Individual country LP at $h = 3$

Country	$\hat{\beta}_{QE}^3$	(s.e.)	$\hat{\beta}_{QT}^3$	(s.e.)
Italy	-0.018	(0.096)	+0.063***	(0.019)
Spain	+0.019	(0.053)	+0.028*	(0.014)
Portugal	+0.011	(0.137)	+0.017	(0.022)
France	+0.009	(0.028)	+0.010	(0.007)

- ▶ QT effect driven by **peripheral economies**: Italy strongest (+0.063***)
- ▶ France almost entirely insulated: near-core status
- ▶ $\Rightarrow$ QT operates through **fragmentation risk**, not uniform repricing

Results: Equity Risk Premia

Baseline specification (QE_Full_shock)

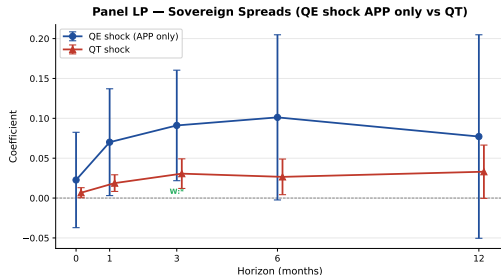
h	$\hat{\beta}_{QE}^h$	$\hat{\beta}_{QT}^h$	Wald p
0	-0.007	-0.003**	0.551
1	-0.013*	-0.007***	0.427
3	-0.003	-0.014***	0.248
6	-0.012	-0.011**	0.915
12	+0.012	- 0.013 **	0.015 **

Key findings:

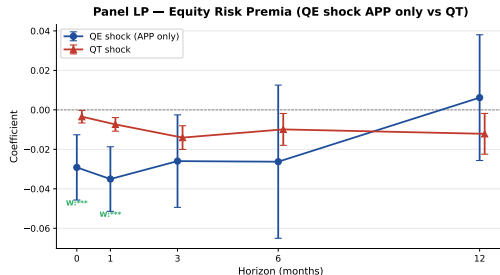
- ▶ QT: **persistent** depression of equity risk premia, significant at all horizons
- ▶ QE: reverts to zero (even turns positive) by $h = 12$
- ▶ Wald rejects symmetry at $h = 12$ ($p = 0.015$): QT $\neq$ QE in persistence

Impulse Response Functions

Sovereign Spreads



Equity Risk Premia



Blue = QE shock, Red = QT shock. 95% Newey-West confidence bands. W:* denotes Wald rejection of symmetry.

Equity-Duration Heterogeneity: Growth vs. Value

Test: interact shocks with sector duration score $D_i \in \{0, 0.5, 1\}$

$$\sum_{j=0}^h \text{ERP}_{t+j}^s = \alpha_i^h + \beta_{QE}^h QE_t + \beta_{QT}^h QT_t + \delta_{QE}^h (QE_t \times D_i) + \delta_{QT}^h (QT_t \times D_i) + \dots$$

Tech = 1 (growth), Banks/OilG = 0.5, Utilities = 0 (value)

h	$\hat{\delta}_{QE}^h$	(s.e.)	$\hat{\delta}_{QT}^h$	(s.e.)
0	+0.001	(0.015)	-0.000	(0.003)
3	+0.009	(0.015)	-0.004	(0.003)
12	+0.024	(0.025)	+0.003	(0.011)

Result: No significant duration heterogeneity at any horizon

- ▶ QT affects **all sectors uniformly** $\Rightarrow$ **risk-taking channel**, not portfolio-balance
- ▶ Robust to restricting to Tech vs. Utilities only

Sensitivity: Restricting QE to the Formal APP (Post-2015)

Baseline uses QE_Full_shock (2009-2022, 98 obs $\neq$ 0). **Alternative:** QE_shock (2015-2022, 52 obs $\neq$ 0).

Sovereign Spreads

h	$\hat{\beta}_{QE}^h$	$\hat{\beta}_{QT}^h$	Wald
0	+0.023	+0.007**	
1	+0.070**	+0.019***	
3	+0.091**	+0.031***	0.075*
12	+0.077	+0.033*	

Equity Risk Premia

h	$\hat{\beta}_{QE}^h$	$\hat{\beta}_{QT}^h$	Wald
0	-0.029***	-0.004**	0.002***
1	-0.035***	-0.007***	0.001***
3	-0.026**	-0.014***	
12	+0.006	-0.012**	

Two complementary asymmetries emerge:

- **Magnitude:** QE (APP) $\gg$ QT at short horizons (Wald $p < 0.01$ on ERP)
- **Persistence:** QE reverts by $h = 12$, QT remains significant

- ▶ **Press-conference shocks:** re-estimated with conference-window OIS surprises
 - No significant effects at any horizon
 - Conference shocks $4\times$ noisier ($\sigma_{conf} \approx 4 \times \sigma_{release}$) and uncorrelated with release shocks ($\rho = -0.006$)
 - Confirms: balance-sheet decisions are communicated in the press release
- ▶ **Individual entity regressions:**
 - Peripheral countries (IT, SP) drive sovereign spread results
 - Banks and Technology exhibit the largest and most persistent equity risk premia compression under QT (at $h=1$ & $h=3$)
 - Oil & Gas structurally decoupled from ECB balance-sheet surprises
- ▶ **QT coefficients stable** across all specifications (baseline, APP-only, conference)

Limitations

1. **QT/rate-hike overlap:** QT begins in June 2022, concurrent with sharp conventional tightening. MP_t^{short} and policy rate control for this, but residual contamination is possible.
2. **Limited QT sample:** only 27 monthly observations $\neq 0$ in the QT regime $\Rightarrow$ long-horizon estimates should be interpreted with caution.
3. **No direct risk-aversion control:** If QT coincides with elevated uncertainty, coefficients may overstate the causal effect.

Conclusion

QT is not simply QE in reverse.

1. **Sovereign spreads:** QT generates persistent widening (up to 12 months), concentrated in peripheral economies. QE compresses spreads only at impact.
2. **Equity risk premia:** QT depresses cumulative excess returns persistently; QE effects fully revert. Wald test rejects long-run symmetry ($p = 0.015$).
3. **No duration heterogeneity:** all equity sectors respond uniformly $\Rightarrow$ risk-taking channel, not portfolio-balance.

Policy implication

The effective financial tightening per unit of balance-sheet reduction may exceed what historical QE episodes predict. So the ECB should be cautious: going slowly and communicating clearly matters a lot when unwinding.

Thank you

Questions?

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